

AWS SAP-C02 - STICKY NOTES CHEAT SHEET

Fast recall • exam traps • architecture decisions • weak-area reminders

1. GOLDEN EXAM RULES

- > Read the LAST sentence first: what exactly is being asked - BEST, MOST secure, LEAST cost, MINIMUM ops, TWO/THREE?
- > MOST secure -> least privilege + private access + temporary credentials + encryption + centralized guardrails.
- > LEAST operational overhead -> native managed/serverless service normally beats custom EC2/scripts.
- > MOST cost-effective -> first satisfy availability/RPO/RTO/security, then optimize cost.
- > Highly available -> usually Multi-AZ. Disaster recovery
- > usually cross-Region.

1. GOLDEN EXAM RULES (cont.)

- > Many AWS accounts -> think Organizations, delegated admin, StackSets, SCP, RAM.
- > Prevent action even for AdministratorAccess -> SCP explicit Deny.
- > Temporary access -> IAM role + STS; avoid IAM users/access keys where roles work.
- TRAP: "Can work" is not the same as "BEST answer".
- Fast elimination checklist
 - Does it meet every stated requirement?
 - Does it require an application rewrite when the question says minimal/no changes?
 - Is there a managed AWS service that replaces custom infrastructure?

1. GOLDEN EXAM RULES (cont.)

- Does it introduce public internet when private connectivity is required?
- Does it create broad permissions when the question says least privilege?

2. IAM + ORGANIZATIONS CORE

- IAM / SCP / Boundary
- > Identity policy -> what this user/role can do.
 - > Resource policy -> who can access this resource.
 - > Permissions boundary -> maximum permissions for an IAM principal.
 - > SCP -> maximum permission guardrail for accounts/OUs; SCP does NOT grant permissions.
 - TRAP: Explicit Deny wins. AdministratorAccess cannot override an SCP Deny.
 - > Current + future accounts in an OU need same restriction -> attach SCP/policy at OU level.
 - Cross-account AssumeRole

2. IAM + ORGANIZATIONS CORE (cont.)

- > Target account role trust policy must trust source principal/account.
- > Source identity must have sts:AssumeRole permission.
- > STS gives temporary credentials - preferred over long-term access keys.
- External ID
 - > Third-party SaaS assuming customer roles -> External ID -> protects against confused deputy problem.
- Role chaining
 - > User/federation -> Role can use target role MaxSessionDuration (up to configured limit).
 - > Role -> AssumeRole -> another Role = role chaining
 - > second role session max 1 hour.

2. IAM + ORGANIZATIONS CORE (cont.)

- TRAP: A target role configured for 12 hours does NOT remove the 1-hour chaining limit.

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3. CROSS-ACCOUNT KMS - HIGH PRIORITY

-> Encrypted cross-account resource = TWO permission layers: resource access + KMS key access.
-> S3 SSE-KMS cross-account -> S3 GetObject permission + KMS Decrypt permission.
-> Encrypted EBS snapshot sharing -> snapshot share permission + customer-managed KMS key permission.
-> Cross-account S3 replication with KMS -> replication rule + IAM replication role + source/destination KMS permissions + destination key configuration.
TRAP: AWS-managed KMS keys are a common trap for cross-account sharing scenarios; customer-

3. CROSS-ACCOUNT KMS - HIGH PRIORITY (cont.)

managed keys give policy control.

4. MULTI-ACCOUNT SECURITY OPERATING MODEL

-> GuardDuty across Organizations -> delegated administrator + auto-enrollment.
-> Security Hub organization-wide -> delegated administrator + central finding aggregation.
-> CloudTrail organization trail -> central Log Archive account/S3 bucket.
-> New accounts automatically receive resources -> CloudFormation StackSets service-managed permissions targeting OUs.
-> Central networking account owns VPC/subnets, app accounts deploy resources there -> VPC Sharing + AWS RAM.

5. NETWORKING DECISION MAP

-> Two VPCs, simple direct private connectivity -> VPC Peering.
-> Many VPCs / hub-and-spoke / centralized routing -> Transit Gateway.
-> Expose one private service to many VPCs/accounts -> PrivateLink.
-> Share actual subnets across accounts -> VPC Sharing + RAM.
-> Dedicated on-prem <-> AWS private circuit -> Direct Connect.
-> Encrypted tunnel over internet -> Site-to-Site VPN.
PrivateLink
-> Provider: NLB -> Endpoint Service. Consumer: Interface Endpoint.

5. NETWORKING DECISION MAP (cont.)

-> Great when CIDRs overlap and consumers must access only one service, not entire provider VPC.
-> Friendly private hostname instead of vpce-* DNS -> endpoint service private DNS + verification + private DNS enabled.
Transit Gateway + inspection
-> TGW route tables control segmentation/propagation: Prod and Dev can be isolated while both reach Shared Services.
-> Stateful firewall + asymmetric return path -> enable TGW appliance mode on inspection VPC attachment.
Gateway vs Interface endpoints

5. NETWORKING DECISION MAP (cont.)

-> S3/DynamoDB Gateway Endpoint -> route-table based, no security group, endpoint policy, low/no endpoint hourly cost.
-> Interface Endpoint -> PrivateLink ENIs/private IPs, security groups, endpoint policy, hourly/data cost.
-> Prevent S3 exfiltration through gateway endpoint -> restrictive endpoint policy + bucket/resource policies.
TRAP: Gateway endpoints do NOT use security groups.

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6. HYBRID DNS

- > AWS -> on-prem DNS = Route 53 Resolver OUTBOUND endpoint + forwarding rules.
- > On-prem -> AWS private hosted zones = Resolver INBOUND endpoint.
- > Many accounts/VPCs -> centralize Resolver endpoints/rules and share rules with RAM.

7. DIRECT CONNECT

- > Multiple VIFs on one physical DX connection != physical redundancy.
- > Two connections at same DX location -> protects device/connection failure, not site failure.
- > Highest resilience -> redundant DX connections on separate devices at separate DX locations + BGP.
- TRAP: If question says survive entire DX location failure, same-location redundancy is insufficient.

8. ROUTE 53 POLICIES

- > Primary/secondary -> Failover.
- > Traffic percentages/canary -> Weighted.
- > Lowest network latency -> Latency-based.
- > User geographic location/country -> Geolocation.
- > Resource geographic position + bias -> Geoproximity.
- > Return multiple healthy records -> Multi-value answer.

9. GLOBAL ACCELERATOR vs CLOUDFRONT

- > Global Accelerator -> static anycast IPs, TCP/UDP, AWS global backbone, fast endpoint failover, no DNS-cache dependency.
- > CloudFront -> HTTP/HTTPS CDN, edge caching, content acceleration, signed URLs/cookies, origin failover.
- TRAP: Fixed global IPs + TCP + no DNS failover dependency -> Global Accelerator, not CloudFront.

10. S3 STORAGE + DATA PROTECTION

Storage classes

- > Frequent access -> S3 Standard.
- > Unknown/changing access -> Intelligent-Tiering.
- > Infrequent, millisecond retrieval -> Standard-IA.
- > Single-AZ acceptable -> One Zone-IA.
- > Archive with immediate-ish access -> Glacier Instant Retrieval.
- > Archive with minutes/hours retrieval -> Glacier Flexible Retrieval.
- > Longest-term cheapest archive -> Glacier Deep Archive.
- Object Lock
- > Governance mode -> privileged principal with bypass permission CAN bypass retention.

10. S3 STORAGE + DATA PROTECTION (cont.)

- > Compliance mode -> protected version cannot be deleted / retention cannot be shortened during retention period.
- > WORM / regulator / even admin cannot delete -> Object Lock Compliance mode.
- > Retention + cheaper tiers + eventual deletion -> Object Lock + Lifecycle policy.
- Replication
- > New objects -> S3 Replication (CRR/SRR depending Regions).
- > Existing objects -> S3 Batch Replication.
- > New + existing -> replication rule + Batch Replication.

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10. S3 STORAGE + DATA PROTECTION (cont.)

-> Cross-account destination ownership required -> configure replica ownership / Object Ownership appropriately.
Multi-Region Access Point
-> Multiple regional S3 buckets + one global S3 endpoint + automatic routing -> S3 Multi-Region Access Point.
TRAP: Transfer Acceleration accelerates transfers to a bucket; it does not create one global multi-bucket access endpoint.

11. CLOUDFRONT + PRIVATE S3

-> Private S3 origin -> CloudFront Origin Access Control (OAC) + bucket policy; keep Block Public Access on.
-> Temporary paid/private access -> signed URL or signed cookie.
-> One/few objects -> signed URL. Multiple restricted objects/path -> signed cookies may fit better.
-> Primary/secondary origins with HTTP failure status codes -> CloudFront Origin Group.

12. EFS / FSx / STORAGE GATEWAY

-> Linux shared NFS -> EFS.
-> Windows SMB + AD + Windows ACLs -> FSx for Windows File Server.
-> HPC/high-performance parallel filesystem -> FSx for Lustre.
-> NetApp enterprise features -> FSx for ONTAP.
-> OpenZFS semantics/features -> FSx for OpenZFS.
-> On-prem app must keep SMB/NFS while authoritative data is in S3 + local cache -> Storage Gateway File Gateway.
-> Block/iSCSI hybrid storage -> Volume Gateway. Virtual tape workflows -> Tape Gateway.

13. DATASYNC + SNOW FAMILY

-> DataSync -> automated online bulk/incremental file data transfer: NFS/SMB/S3/EFS/FSx patterns.
TRAP: DataSync does NOT convert Oracle/SQL Server databases to PostgreSQL/Aurora.
-> Huge initial dataset + bandwidth constrained -> Snowball Edge / Snow Family for offline seeding.
-> After bulk seed, recurring daily increments -> DataSync.

14. RDS / AURORA DECISION RULES

-> RDS Multi-AZ -> high availability / failover.
-> Read Replica -> read scaling/reporting.
TRAP: Traditional RDS Multi-AZ standby is not your reporting endpoint.
-> Cross-Region read replica -> DR copy/read capability + promotion option.
Aurora Global Database
-> Low-latency cross-Region replication, seconds-level RPO, fast secondary promotion -> Aurora Global Database.
-> Secondary Region serves local reads. Eligible occasional writes can use Global Database write forwarding to primary.

14. RDS / AURORA DECISION RULES (cont.)

TRAP: RDS Proxy does NOT make a secondary Aurora cluster writable.
RDS Proxy
-> Lambda + RDS/Aurora + connection storm/exhaustion
-> RDS Proxy.
-> RDS Proxy -> connection pooling/reuse, improved connection management; credentials commonly via Secrets Manager.
Aurora Serverless v2
-> Relational + highly variable demand + automatic granular scaling + low ops -> Aurora Serverless v2.

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15. DYNAMODB

-> Massive scale + low-latency key-value/document + serverless -> DynamoDB.
-> Multi-Region multi-active reads/writes -> DynamoDB Global Tables.
-> Few partition keys generate huge traffic -> hot partition -> improve key cardinality / write sharding.
-> Write sharding example: customer#0, customer#1, customer#2; query all shards when reading customer history.
TRAP: Adding a GSI with the SAME hot partition key does not solve skew.

16. MIGRATION + MODERNIZATION TOOL MAP

-> Server/VM rehost -> AWS Application Migration Service (MGN).
-> Database data migration + CDC -> AWS DMS.
-> Heterogeneous DB schema/code conversion -> DMS Schema Conversion / AWS SCT.
-> File transfer/sync -> DataSync.
-> PB-scale offline transfer -> Snow Family.
-> Discover servers/utilization/dependencies -> Application Discovery Service / migration discovery tooling.
-> Track migrations centrally -> Migration Hub.
Heterogeneous DB pattern

16. MIGRATION + MODERNIZATION TOOL MAP (cont.)

-> Oracle/SQL Server -> Aurora PostgreSQL = schema/code conversion + DMS full load + CDC.
-> Full load moves baseline. CDC keeps target nearly synchronized until cutover.
TRAP: MGN replicates servers; it does not transform Oracle into PostgreSQL.
TRAP: Native SQL Server backup cannot be restored directly into Aurora PostgreSQL.

17. DISASTER RECOVERY STRATEGIES

-> Backup & Restore -> cheapest, highest RPO/RTO.
-> Pilot Light -> core critical services/data running; scale/build rest during disaster.
-> Warm Standby -> functional reduced-capacity environment already running; scale up during disaster.
-> Multi-Site / Active-Active -> both Regions serving, highest cost, lowest RPO/RTO.
-> Secondary app stack exists but reduced + global DB + scale on failover -> Warm Standby.

18. SQS / SNS / EVENTBRIDGE / KINESIS

-> SQS -> queue, buffering, decoupling, async work.
-> SNS -> push pub/sub fan-out.
-> EventBridge -> event routing/filtering/integration across producers/consumers/accounts.
-> Kinesis Data Streams -> high-throughput ordered streaming per partition key + replay + multiple consumers.
SQS Standard vs FIFO
-> Standard -> very high throughput, at-least-once delivery, best-effort ordering.
-> FIFO -> ordering + MessageGroupId + deduplication features.

18. SQS / SNS / EVENTBRIDGE / KINESIS (cont.)

-> Same MessageGroupId = ordered serial flow. Different groups = parallelism.
Lambda + SQS
-> One malformed message makes whole batch retry -> enable partial batch responses and return failed message identifiers.
-> Repeated poison message -> SQS DLQ + appropriate maxReceiveCount.
Idempotency - exam favorite
-> Exactly-once BUSINESS effect -> application-level idempotency using unique transaction/request ID + durable check.
TRAP: FIFO/deduplication does not replace idempotent payment/order logic.

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19. STEP FUNCTIONS

- > Multi-step workflow + state + retry + catch + branching + compensation -> Step Functions.
- > Payment succeeded but inventory failed -> Catch -> compensation/refund step.
- > EventBridge routes events; SQS buffers; Step Functions orchestrates workflow.

20. AUTO SCALING + SPOT

- > SQS worker scaling -> custom metric "backlog per worker" ~ queue depth / active workers; use target tracking.
- > Spot best for stateless, retryable, checkpointed, fault-tolerant workloads.
- > Resilient Spot -> multiple instance types + multiple AZs + price-capacity-optimized allocation strategy.
- TRAP: One instance type + one AZ = narrow Spot pool and more interruption risk.

21. ALB vs NLB

- > ALB -> Layer 7 HTTP/HTTPS, host/path routing, WebSocket/gRPC use cases.
- > NLB -> Layer 4 TCP/UDP/TLS, extreme performance, static IP support, PrivateLink provider backend.
- > Path/host based routing -> ALB. PrivateLink endpoint service -> NLB.

22. SECURITY SERVICE MAP

- > GuardDuty -> threat detection.
- > Security Hub -> central security findings/posture aggregation.
- > Inspector -> vulnerability management for supported workloads/images/functions.
- > Macie -> sensitive data discovery in S3.
- > CloudTrail -> who did what AWS API action.
- > CloudWatch -> metrics, logs, alarms, observability.
- > AWS Config -> configuration state/history/compliance.
- > Secrets Manager -> secrets + automatic rotation use cases.

22. SECURITY SERVICE MAP (cont.)

- > Systems Manager Parameter Store -> configuration/parameters and simpler secret storage use cases.

23. EKS / WORKLOAD IDENTITY

- > Specific Kubernetes service account needs AWS permissions -> pod/service-account IAM identity such as IRSA; do not give whole worker-node role broad access.
- > Cross-account S3 from EKS -> scoped pod identity/role + target bucket/KMS permissions as needed.

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24. DATA LAKE GOVERNANCE

-> S3 stores data. Glue Data Catalog stores metadata. Athena queries. Lake Formation governs permissions.
-> Cross-account centralized data lake + table/column permissions -> Lake Formation.

25. CLOUDFORMATION STACKSETS

-> Same IAM/Config/CloudWatch resources across 200 accounts and Regions -> StackSets.
-> Organizations integration + service-managed permissions + OU targets -> automatic deployment to new accounts.

26. COST OPTIMIZATION CLUES

-> Steady predictable compute -> Savings Plans / Reserved pricing approaches.
-> Interruptible fault-tolerant compute -> Spot.
-> Unknown S3 access pattern -> Intelligent-Tiering.
-> Archive -> Glacier families based on retrieval requirement.
-> Variable event-driven compute -> Lambda/serverless.
-> Variable relational database -> Aurora Serverless v2.
TRAP: Do not pick cheapest option if it violates RPO/RTO/security/latency requirements.

27. YOUR MOCK #1 WEAK AREAS - MEMORIZE

Cross-account KMS
-> Resource permission + KMS permission. Think two doors.
Migration service selection
-> MGN=server; DMS=data; Schema Conversion/SCT=schema+code; DataSync=files.
Global routing
-> Fixed IP + TCP/UDP + no DNS dependency -> Global Accelerator.
-> One global S3 endpoint across regional buckets -> Multi-Region Access Point.
S3 retention
-> Even privileged admin cannot delete during retention -> Object Lock Compliance.
DynamoDB
-> Hot key/partition -> write sharding / better partition key.
IAM

27. YOUR MOCK #1 WEAK AREAS - MEMORIZE (cont.)

-> Role -> Role = max 1 hour chaining session. Exactly-once business effect
-> Idempotency required; messaging guarantees alone are not enough.
Aurora
-> Secondary local reads + occasional writes -> write forwarding; RDS Proxy is only connection management.

28. "SEE THIS -> THINK THIS" FINAL RECALL

Admin must be blocked -> SCP explicit Deny
Third-party AssumeRole -> External ID
Cross-account encrypted -> Resource permission + KMS permission
Role assumes role -> 1-hour chaining
Many VPCs -> Transit Gateway
One private service -> PrivateLink
Share subnet -> RAM / VPC Sharing
TGW + stateful firewall -> Appliance Mode
Fixed global IP -> Global Accelerator
Global web cache -> CloudFront
Private S3 via CloudFront -> OAC
S3 immutable -> Object Lock Compliance
Existing S3 replication -> Batch Replication

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28. "SEE THIS -> THINK THIS" FINAL RECALL (cont.)

Global S3 endpoint -> Multi-Region Access Point
Windows SMB -> FSx Windows
Linux NFS -> EFS
Hybrid SMB/NFS + S3 -> File Gateway
Lambda DB connection storm -> RDS Proxy
Aurora cross-Region -> Global Database
Aurora secondary writes -> Write Forwarding
RDS HA -> Multi-AZ
RDS reads -> Read Replica
DynamoDB global writes -> Global Tables
DynamoDB hot partition -> Write Sharding
VM rehost -> MGN
DB data migration -> DMS
DB schema conversion -> SCT / DMS Schema Conversion
File migration -> DataSync
Huge offline data -> Snowball

28. "SEE THIS -> THINK THIS" FINAL RECALL (cont.)

Discover dependencies -> Application Discovery
Ordered stream + replay -> Kinesis
Queue -> SQS
Fan-out -> SNS
Event routing -> EventBridge
Workflow -> Step Functions
Poison messages -> DLQ
Bad message in Lambda/SQS batch -> Partial Batch Response
Duplicate payment -> Idempotency
Specific EKS workload IAM -> IRSA / pod identity
Multi-account deployment -> StackSets
Data lake fine-grained permissions -> Lake Formation
Threat detection -> GuardDuty
Central findings -> Security Hub
API audit -> CloudTrail

28. "SEE THIS -> THINK THIS" FINAL RECALL (cont.)

Config compliance -> AWS Config

29. LAST-MINUTE EXAM METHOD

-> Step 1: Identify hard requirements (RPO/RTO, protocol, public/private, cross-account, Region/AZ, latency, cost).
-> Step 2: Eliminate any answer violating even one hard requirement.
-> Step 3: Among remaining answers, prefer AWS-managed/native integration and least operational overhead.
-> Step 4: For security questions, prefer temporary credentials, least privilege, resource + key permissions, organization guardrails.

29. LAST-MINUTE EXAM METHOD (cont.)

-> Step 5: For "MOST cost-effective", choose the cheapest architecture that still meets every technical requirement.
-> Step 6: Never leave an answer blank; there is no benefit to not guessing when time expires.
TRAP: Watch the words BEST, MOST, LEAST, MINIMUM, TWO, THREE. Missing the qualifier can lose an easy mark.

30. FINAL MEMORY MANTRA

"Meet ALL requirements first. Then choose the most AWS-native, least operational, least-privilege design."
Mock #1 result: $63/75 = 84\%$. Your fundamentals are strong; turn the weak-area patterns into instant recall.